

GOOGLE MAPS MERCHANT SCRAPING TECHNICAL PLAN

Indonesia Total Addressable Market (TAM) Expansion

Financial Services – Data Science Team | April 2025

1. Objective

This document outlines the technical plan to scrape merchant data across all districts in Indonesia using the Google Maps Places API (Text Search – New). The goal is to build a comprehensive merchant database to support TAM analysis for the business team's product expansion strategy.

Key outcomes expected from this scrape:

- Full merchant coverage across Indonesia by district (Kabupaten/Kota or Kecamatan level)
- Categorized merchant data covering FMCG and F&B verticals
- Geocoded locations (lat/lng) with ratings, opening hours, phone numbers, and addresses
- Clean, deduplicated merchant dataset filtered to each district's GeoJSON boundary

2. API Selection

The Google Maps Places API (Text Search – New) is selected for this project for the following reasons:

- Supports keyword + location queries (e.g. "minimarket", "restaurant") — ideal for category-based searches
- Supports locationRestriction to roughly constrain results to a bounding box derived from the district GeoJSON
- Returns up to 20 results per page with nextPageToken-based pagination, up to 10 pages (200 results maximum per query)
- Pro SKU provides all required fields: display name, rating, opening hours, coordinates, phone, and website

API Documentation: <https://developers.google.com/maps/documentation/places/web-service/text-search>

3. Data Fields: Essentials vs. Pro SKU

Billing is determined by the highest-tier field requested. Since this project requires display name, rating, and opening hours, the Pro SKU is mandatory for all queries.

Field	Essentials	Pro (Used)
Place ID	✓	✓
Lat / Lng	✓	✓
Formatted Address	✓	✓
Business Type	✓	✓
Viewport / BBox	✓	✓
Display Name (text)	✗	✓
Rating	✗	✓
Opening Hours	✗	✓
Phone Number	✗	✓
Website	✗	✓
Price Level	✗	✓
Photos	✗	✓

4. Search Category Taxonomy

A total of 20 search categories are defined across two verticals: FMCG and F&B. Each category maps to either a Google Place Type (used in the includedType parameter) or a Keyword query (used in the textQuery parameter).

4.1 FMCG Categories (10)

FMCG categories target outlets selling fast-moving consumer goods — from modern minimarkets and supermarkets to traditional toko kelontong and specialty stores.

#	Query / Type	Query Method	Description
1	supermarket	Place Type	Modern supermarkets (Indomaret+)
2	convenience_store	Place Type	Minimarkets & convenience stores
3	drugstore	Place Type	Apotek / pharmacy chains
4	hardware_store	Place Type	Building & household supplies
5	beauty_supply	Place Type	Cosmetics & personal care stores
6	pet_store	Place Type	Pet food & supplies
7	liquor_store	Place Type	Beverages & alcohol retail
8	toko kelontong	Keyword	Traditional grocery (warung)
9	minimarket	Keyword	Small modern retail stores
10	toko sembako	Keyword	Staple goods traditional stores

4.2 F&B Categories (10)

F&B categories cover all food and beverage outlets from full-service restaurants to traditional warung makan and coffee stalls. Both Place Types and keyword queries are used to maximize coverage of formal and informal establishments.

#	Query / Type	Query Method	Description
11	restaurant	Place Type	Full-service restaurants
12	cafe	Place Type	Coffee shops & cafes
13	bakery	Place Type	Bakeries & bread shops
14	fast_food_restaurant	Place Type	Fast food chains
15	meal_takeaway	Place Type	Take-away & delivery kitchens
16	bar	Place Type	Bars & nightlife F&B
17	ice_cream_shop	Place Type	Dessert & ice cream outlets
18	warung makan	Keyword	Traditional Indonesian eateries
19	kedai kopi	Keyword	Traditional & modern coffee stalls
20	food court	Keyword	Multi-tenant food courts

Note: Place Type queries are passed using the includedType parameter for higher precision. Keyword queries use the textQuery field directly and are post-filtered using the GeoJSON polygon.

5. District Coverage

Indonesia's administrative geography is divided into three levels. This project will execute searches at the Kabupaten/Kota level as the primary scope, with an optional Kecamatan drill-down for high-priority cities.

Level	Count	Recommendation
Kecamatan	~7,277	Recommended for Phase 1 – deep-dive into high-priority cities
Kelurahan / Desa	~83,000+	Not recommended – extremely high cost and diminishing data quality returns

6. GeoJSON Boundary Handling

Google Maps Text Search does not natively accept GeoJSON polygons. A two-layer approach is used to enforce district boundaries:

Layer 1 – Query-time Bounding Box

Each district's GeoJSON polygon is converted to its bounding box (min/max lat/lng). This is passed as the locationRestriction parameter (rectangle) to roughly constrain the search area.

Layer 2 – Post-query Point-in-Polygon Filter

Every merchant returned by the API is checked against the actual GeoJSON polygon using Shapely's point-in-polygon function. Any merchant whose coordinates fall outside the district polygon is discarded.

Python implementation:

```
from shapely.geometry import shape, Point
polygon = shape(geojson['geometry'])
filtered = [m for m in results if polygon.contains(Point(m['lng'], m['lat']))]
```

7. Pagination Strategy

- Each API call returns up to 20 results per page
- The API supports up to 10 pages per query (200 results maximum)
- Pagination is nextPageToken-driven — only fetch the next page if the token exists
- Average pages assumed for cost estimation: 5 pages per query
- Dense urban areas (Jakarta, Surabaya, Bandung) may reach 10 pages for popular categories
- Rural kecamatan will likely return 1–2 pages for most categories, reducing actual cost

Deduplication is applied after all pages are collected using the Google Place ID as the unique key. Duplicate place_id values across category queries in the same district are merged to retain a single record.

8. Pricing Estimation

8.1 Assumptions

Parameter	Value
API SKU	Places API Text Search – Pro
Price (Tier 1, 0–100K)	\$32.00 per 1,000 requests
Price (Tier 2, 100K–500K)	\$25.60 per 1,000 requests
Price (Tier 3, 500K–1M)	\$19.20 per 1,000 requests
Price (Tier 4, 1M–5M)	\$9.60 per 1,000 requests
Free Tier (Pro SKU)	5,000 requests / month
Total Search Categories	20 (10 FMCG + 10 F&B)
Average Pages per Query	5 pages (100 results per query)
Results per Page	20 results (each page = 1 billable request)

8.2 Cost Calculation Formula

Total Requests = District Count × 20 Categories × 5 Avg Pages
Billable Requests = max(0, Total Requests - 5,000 free)

8.3 Scenario Cost Summary

District Level	Districts	Total Requests	Billable Req.	Est. Total Cost (USD)
Kabupaten / Kota	514	51,400	46,400	\$1,484.80
Kecamatan (with volume disc.)	7,277	727,700	722,700	\$17,651.84

* Kecamatan cost applies volume discount tiers: \$32/K for first 100K billable, \$25.60/K for 100K–500K, \$19.20/K for 500K–1M requests.

8.4 Kecamatan Tier Breakdown

Volume Tier	Requests	Rate / 1K	Subtotal
Free tier	5,000	\$0.00	\$0.00
Tier 1: 0 – 100,000	95,000	\$32.00	\$3,040.00
Tier 2: 100,001 – 500,000	400,000	\$25.60	\$10,240.00
Tier 3: 500,001 – 727,700	227,700	\$19.20	\$4,371.84
TOTAL	727,700	–	\$17,651.84

9. Implementation Notes

Rate Limiting

- Implement request throttling at 5–10 requests per second to avoid quota errors
- Use exponential backoff on HTTP 429 (Too Many Requests) errors

Deduplication

- Use place_id as the unique key — a merchant appearing in multiple category queries in the same district should only be stored once
- Merge the types array to capture all applicable categories per merchant

Storage

- Store raw responses as JSON per district/category for auditability
- Write a cleaned, flattened CSV/BigQuery table with one row per unique merchant

GeoJSON Sources

- Indonesia administrative boundaries are available from BIG (Badan Informasi Geospasial) and the GADM database
- Ensure GeoJSON polygons use WGS84 (EPSG:4326) — same coordinate system as Google Maps API

Cost Control

- Set a Google Cloud billing budget alert at \$500 intervals
- Run a pilot on 5 kabupaten before full execution to validate data quality and pagination depth
- Monitor average pages-per-query in the pilot — if actual average is below 5, revise cost estimates downward

10. Summary

Item	Detail
API	Google Maps Places API – Text Search (New)
SKU	Pro – \$32.00 / 1,000 requests (volume discounts apply)
Categories	20 (10 FMCG + 10 F&B)
Avg Pages / Query	5 pages (100 results)
Full Kecamatan Cost	~\$17,652 for all 7,277 kecamatan (with volume discounts)
Boundary Method	locationRestriction (bounding box) + Shapely point-in-polygon post-filter
Deduplication Key	Google Place ID
Recommended Start	Phase 1 – Kabupaten/Kota nationwide sweep, then prioritized kecamatan drill-down